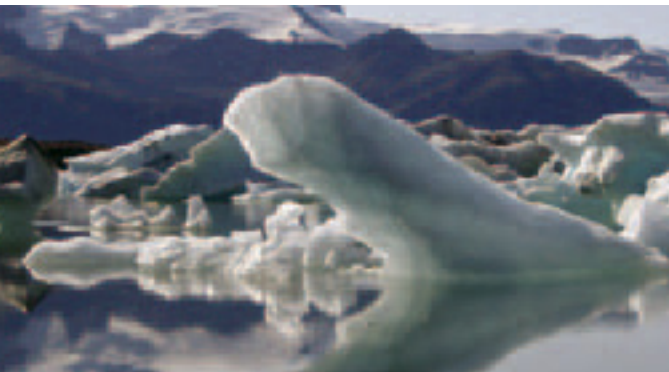
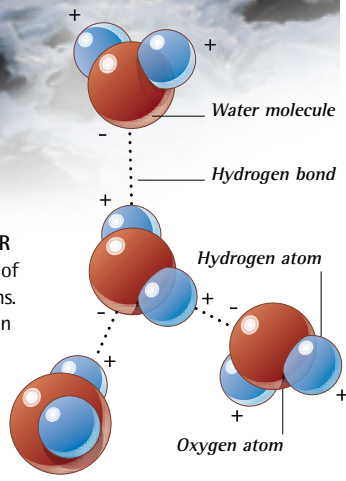


OCEAN WATER

Most of the water that fills the oceans originally came from inside the Earth. It was expelled from volcanoes as water vapor early in the history of the planet, and due to ideal conditions of temperature and gravity, this vapor eventually condensed into vast pools of liquid water that we call oceans. This ocean water is far from pure, since it contains many dissolved gases and minerals, including those that make it salty. It also contains the chemicals that form living molecules, making it likely that the oceans were the cradle of life on Earth.

▲ THE NATURE OF WATER

Water molecules are made of hydrogen and oxygen atoms. Weak electrostatic hydrogen bonds make the water molecules cling together as a liquid. At slightly higher temperatures, they drift apart to form a gas called water vapor. At lower temperatures, they lock into solid ice. Water is unique because these three states can occur in the same place, at the same time.



◀ ICE

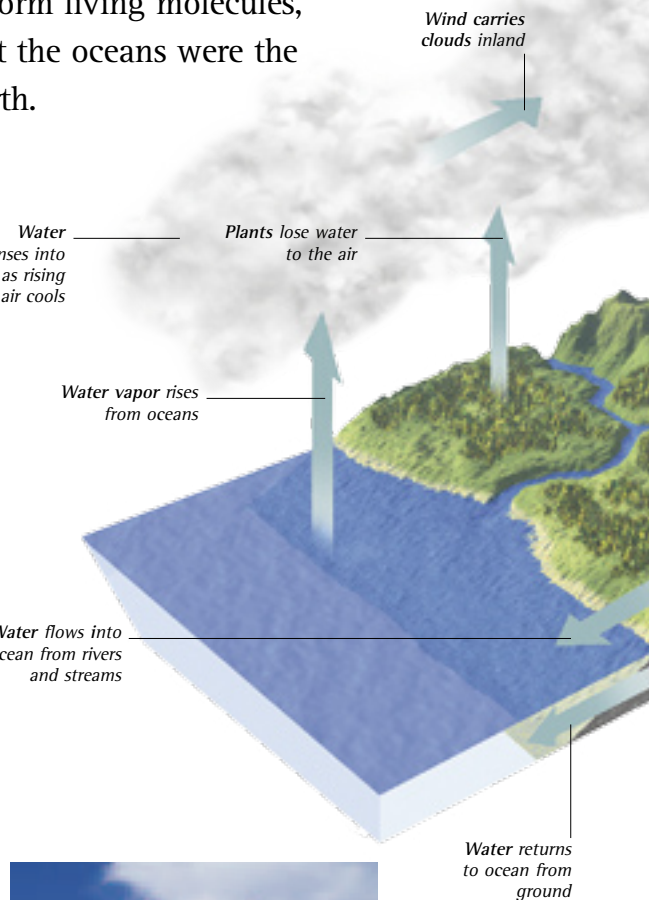
When the temperature of pure water falls below 32°F (0°C), strong hydrogen bonds lock their molecules together to form solid ice. Salty seawater freezes at a slightly lower temperature of 28.8°F (-1.8°C). Uniquely, the solid, icy form of water is slightly less dense than liquid water, which is why icebergs and ice cubes float.

◀ WATER

At temperatures above its freezing point, ice melts to form water. This is slightly denser and heavier, but the molecules are less tightly bound together. They can move around, and this is why water is a liquid. But the attractive forces between them are still strong, making water cling together in droplets.

◀ CLOUD

At 212°F (100°C), water boils and turns into water vapor, a gas. The water molecules break apart and float freely in the air. But this can also happen at lower temperatures, so water vapor is always rising off the oceans. It is invisible, but if it cools it condenses into droplets that form clouds.



▲ VOLCANIC ORIGINS

Nearly all the water on Earth was probably ejected from volcanoes when Earth's core was forming, more than 4.5 billion years ago. Volcanoes still produce vast amounts of water vapor and other gases, and this mixture would have formed an early atmosphere. The vapor condensed into rain that formed the first oceans. About 4 billion years ago, these may have covered the whole planet.